

Chapter 2 - Modules

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Exercise 1

Since $\gcd(m, n) = 1, \exists u, v \in \mathbb{Z}, \text{ s.t.}$

$$mu + nv = 1$$

Now for $\forall x \otimes y \in (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}),$

$$\begin{aligned} x \otimes y &= (mu + nv)(x \otimes y) \\ &= mu(x \otimes y) + nv(x \otimes y) \\ &= 0 \end{aligned}$$

□

Exercise 2

Consider the exact sequence

$$\mathfrak{a} \xrightarrow{f} A \xrightarrow{g} A/\mathfrak{a} \rightarrow 0$$

By *Proposition 2.18*, we have the following exact sequence

$$\mathfrak{a} \otimes M \xrightarrow{f \otimes 1} A \otimes M \xrightarrow{g \otimes 1} (A/\mathfrak{a}) \otimes M \longrightarrow 0$$

Notice there exist trivial isomorphism $\mathfrak{a} \otimes M \cong \mathfrak{a}M, A \otimes M \cong M$. So we can have the exact sequence

$$\mathfrak{a}M \xrightarrow{\varphi} M \xrightarrow{\psi} (A/\mathfrak{a}) \otimes M \rightarrow 0$$

With $\text{Im } \varphi = \text{Ker } \psi, \text{Im } \psi = (A/\mathfrak{a}) \otimes M$. Which is to say

$$\text{Im } \psi = (A/\mathfrak{a}) \otimes M, \text{Ker } \psi = \mathfrak{a}M$$

So by *Module Isomorphism Theorem*,

$$(A/\mathfrak{a}) \otimes M \cong M/\mathfrak{a}M$$

□

Exercise 3

We follow from the hint: For any finitely generated A -module M_0 , let $\mathfrak{m}$ be the maximal ideal of A and $k = M_0/\mathfrak{m}$ be the field derived from quotient. Now by *Ex-2.2*, $k \otimes_A M_0 \cong M_0/\mathfrak{m}M_0$. So by *Nakayama Lemma*, $k \otimes_A M_0 = 0$ will indicate $M_0 = 0$.

Back to the problem:

$$\begin{aligned} 0 &= k \otimes_A k \otimes_A \otimes_A M \otimes_A N \\ &\cong k \otimes_A (k \otimes_A M) \otimes_A N \\ &\cong k \otimes_A M_k \otimes_A N \\ &\cong M_k \otimes_A (k \otimes_A N) \\ &\cong M_k \otimes_A N_k \end{aligned}$$

Exercise 4

For any A -module N , consider

$$f : N \longrightarrow N'$$

which is injective. Now

$$M \text{ is flat}$$

$\Leftrightarrow$

$$f \otimes 1 : N \otimes M \longrightarrow N' \otimes M \text{ is injective}$$

$\Leftrightarrow$ (Be considering their kernel)

$$\forall i \in I, f \otimes 1 : N \otimes M_i \longrightarrow N' \otimes M_i \text{ is injective}$$

□

Exercise 5

We can view $A, xA, x^2A, \dots$ as a sequence of A -module, and $A[x]$ is their direct sum where multiplication in the A -algebra structure defined in the trivial way.

Exercise 6

Define

$$\varphi : A[x] \otimes_A M \longrightarrow M[x]$$

by mapping

$$p(x) \otimes m \longmapsto p(x) \cdot m$$

Obviously $\text{Ker } \varphi = 0$, $\text{Im } \varphi = M[x]$, so φ is an isomorphism.

Exercise 7

- (i) If $a(x), b(x) \in A[x], a(x)b(x) \in \mathfrak{p}[x]$. Let $a(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0, b(x) = b_n x^n + b_{n-1} x^{n-1} + \dots + b_1 x + b_0$, then

$$(a_n x^n + \dots + a_1 x + a_0)(b_n x^n + \dots + b_1 x + b_0) \in \mathfrak{p}[x]$$

$$a_n b_n \in \mathfrak{p} \Rightarrow a_n \in \mathfrak{p} \text{ or } b_n \in \mathfrak{p}$$

Suppose $a_n \in \mathfrak{p}$, then let $a'(x) = a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0$ and we have

$$a'(x)b(x) \in \mathfrak{p}[x]$$

Repeat the process above and eventually

$$a(x) \in \mathfrak{p}[x] \text{ or } b(x) \in \mathfrak{p}[x]$$

□

- (ii) Not necessarily.

Let $A = k$ be a field and $\mathfrak{m} = (0)$. Obviously $\mathfrak{m}[x] = (0)$ is not a maximal ideal in $k[x]$.

Exercise 8

- (i) Consider the injective A -module homomorphism

$$\varphi : R \longrightarrow R'$$

Since M is flat,

$$\varphi \otimes 1 : R \otimes_A M \longrightarrow R' \otimes_A M$$

is injective, therefore

$$\varphi \otimes 1 \otimes 1 : R \otimes_A M \otimes_A N \longrightarrow R' \otimes_A M \otimes_A N$$

is injective, i.e.

$$\varphi \otimes 1 : R \otimes_A (M \otimes_A N) \longrightarrow R' \otimes_A (M \otimes_A N)$$

is injective. So $M \otimes_A N$ is flat.

(ii) Consider a injective B -module homomorphism

$$\varphi : M \longrightarrow M'$$

Since N is a flat B -module homomorphism,

$$\varphi \otimes_B 1 : M \otimes_B N \longrightarrow M' \otimes_B N$$

is injective. To prove N is a flat A -module, we need to show

$$\varphi \otimes_A 1 : M \otimes_A N \longrightarrow M' \otimes_A N$$

is injective, i.e. $\text{Ker } \varphi \otimes_A 1 = \{0\}$.

Notice $\text{Ker } \varphi \otimes_A 1 \subseteq \text{Ker } \varphi \otimes_B 1 = \{0\}$, we're done. $\square$

Exercise 9

Denote

$$0 \rightarrow M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$$

We know

$$M \cong \text{Coim } g \oplus \text{Ker } g \cong \text{Im } g \oplus \text{Ker } g$$

Be the exactness, $\text{Ker } g = \text{Im } f$, $\text{Im } g = M''$, which are both finitely generated. So M is finitely generated. $\square$

Exercise 10

Since $M/\mathfrak{a}M \longrightarrow N/\mathfrak{a}N$ is surjective,

$$M \longrightarrow M/\mathfrak{a}M \longrightarrow N/\mathfrak{a}N$$

is surjective. Therefore $N = \mathfrak{a}N + u(M)$, and by *Corollary 2.7*, $u(M) = N$. $\square$

Exercise 11

(The ring multiplication here is Cartesian product!!!)

(i) When $A^m \cong A^n$: We use the method in the textbook.

Let $\mathfrak{m}$ be a maximal ideal of A , and denote the isomorphism by

$$\phi : A^m \longrightarrow A^n$$

By checking the kernel and image, we can easily see that

$$1 \otimes_A \phi : (A/\mathfrak{m}) \otimes_A A^m \longrightarrow (A/\mathfrak{m}) \otimes_A A^n$$

is an isomorphism by *Proposition 2.18*. By *Exercise 2.2* we know $(A/\mathfrak{m})^m \cong A^m/\mathfrak{m} \cong (A/\mathfrak{m}) \otimes A^m$, $(A/\mathfrak{m})^n \cong A^n/\mathfrak{m} \cong (A/\mathfrak{m}) \otimes A^n$ are two isomorphic linear space over the field $k = A/\mathfrak{m}$. Hence

$$\text{Dim } k^m = \text{Dim } k^n, m = n$$

(ii) When $\phi : A^m \longrightarrow A^n$ is surjective: Similar to (i), we obtain a surjective k -linear space homomorphism by using *Proposition 2.18*

$$k^m \longrightarrow k^n$$

So $m > n$ is obtained by the property of linear space homomorphisms.

(iii) When $\phi : A^m \longrightarrow A^n$ is injective: Yes.

Note that we're no longer able to use *Proposition 2.18* to convert ring homomorphisms into linear space homomorphisms.

Suppose $m > n$, we can define A^n to be the submodule of A^m by taking the first n terms and extend the range of ϕ to A^m . Now

$$\phi(A^m) \subseteq A^n \subseteq A^m = \mathfrak{a}A^m$$

where $\mathfrak{a} = A$ is viewed as an ideal.

Since A^m is a finitely generated A -module, by *Proposition 2.4*, $\exists a_1, a_2, \dots, a_s \in A$, s.t.

$$\phi^s + a_1\phi^{s-1} + \dots + a_s = 0, a_s \neq 0$$

Specifically,

$$(\phi^s + a_1\phi^{s-1} + \dots + a_{s-1}\phi + a_s\text{Id})(0, \dots, 0, 1) = 0$$

Notice $\forall a \in A^m$, because $m > n$, the last term of $\phi(a)$ must be 0. So

$$(\phi^s + \dots + a_{s-1}\phi)(0, \dots, 0, 1) = (\dots, 0)$$

But

$$a_s\text{Id}(0, \dots, 0, 1) = (\dots, a_s)$$

Hence, it's impossible to have $(\phi^s + a_1\phi^{s-1} + \dots + a_{s-1}\phi + a_s\text{Id})(0, \dots, 0, 1) = 0$ So $m \leq n$.

□

Exercise 12

Let $A^n = \langle e_1, e_2, \dots, e_n \rangle$ where

$$\begin{cases} e_1 = (1, 0, 0, \dots, 0) \\ e_2 = (0, 1, 0, \dots, 0) \\ \vdots \\ e_n = (0, 0, 0, \dots, 1) \end{cases}$$

and choose $u_k \in M$, s.t. $\phi(u_k) = e_k$. We can prove every element in M can be expressed in the sum of element in $\langle u_1, u_2, \dots, u_n \rangle, \text{Ker } \phi$. Also, $\text{Ker } \phi \cap \langle u_1, \dots, u_n \rangle = \{0\}$, so $M = \text{Ker } \phi \oplus \langle u_1, \dots, u_n \rangle$.

Exercise 13

(i) $\text{Ker } g = 0$, so g is injective.

(ii) Define

$$p : N_B \longrightarrow N$$

by

$$b \otimes y \longmapsto by$$

and it's trivial that

$$N_B = \text{Im } g \oplus \text{Ker } p$$

□

Exercise 14

To prove $\mu_i = \mu_j \circ \mu_{ij}$, we only need to understand that in the structure of $M = C/D$, elements are the equivalence classes in C where equivalence relation defined by homomorphisms. The action of μ_{ij} is simply sending one element to another in the same equivalence class.

After one understand this, the problem becomes rather simple.

Exercise 15

(i) This is to say that $\mu : C \longrightarrow C/D$ is surjective, which is obviously correct.

(ii) This is automatically obtained by the definition of D .

Exercise 16

Because any such homomorphism that satisfy

$$\forall i \in I, \alpha_i = \alpha \circ \mu_i$$

is unique determined by the inner structure of M_i, N, M .

Exercise 17

Because in the notation of *Exercise 2.14*, $D = \{0\}$ and therefore

$$\lim_{\rightarrow} M_i = C/D = C = \sum_{i \in I} M_i$$

□

Exercise 18

$$\begin{array}{ccc} M_i & \xrightarrow{\phi_i} & N_i \\ \downarrow \mu_i & & \downarrow \nu_i \\ M & \xrightarrow{\phi} & N \end{array}$$

Because from those $\nu_i \circ \phi_i$ defining the direct limit homomorphism $M_i \rightarrow N$, the unique homomorphism ϕ is obtained by the universal property of direct limit.

Exercise 19

$$\begin{array}{ccccc} M_i & \xrightarrow{\phi_i} & N_i & \xrightarrow{\psi_i} & P_i \\ \downarrow \mu_i & & \downarrow \nu_i & & \downarrow \xi_i \\ M & \xrightarrow{\phi} & N & \xrightarrow{\psi} & P \end{array}$$

By *Exercise 2.15*, each elements in M can be expressed in the form $\mu_i(x_i)$, and $\text{Ker } \psi = \{\nu_i \text{Ker } \psi_i\}_{i \in I}$. Therefore

$$\begin{aligned} \text{Im } \phi &= \{\text{Im } \phi \circ \mu_i\}_{i \in I} \\ &= \{\text{Im } \nu_i \circ \phi_i\}_{i \in I} \\ &= \{\nu_i \text{Im } \phi_i\}_{i \in I} \\ &= \{\nu_i \text{Ker } \psi_i\}_{i \in I} \\ &= \text{Ker } \psi \end{aligned}$$

□

Exercise 20

$$\begin{array}{ccc} M_i \times N & \longrightarrow & M_i \otimes N \\ \downarrow & & \downarrow \\ (\lim_{\rightarrow} M_i) \times N & \longrightarrow & \lim_{\rightarrow} (M_i \otimes N) \end{array}$$